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I Semester M.C.A.(2 Years Course) Degree Examination, August/September 2021
(CBCS Scheme) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA3 : Computer Organization and Architecture

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** from Section – **A** and **any four** from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks : **(5×6=30)**

1. Convert the following :
   i) $7562_{(10)} = _____{}_{(16)}$
   ii) $1110101_{(2)} = _____{}_{(10)}$
   iii) $F3A7_{(16)} = _____{}_{(8)}$
2. Construct the basic logic gates using NOR logic gate.
3. Differentiate CISC and RISC.
4. Explain different instruction formats with an example for each.
5. Explain the limitation of Instruction level parallelism.
6. DMA has Priority over the CPU when both request a memory transfer. Justify your answer.
7. Explain the characteristics of multiprocessors.
8. Explain different types of ROM.

P.T.O.

SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. a) Minimize the following expression using K-Map **5**

   $Y = \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + AB\bar{C}\bar{D} + AB\bar{C}D + A\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D}$

   b) Design a combination logic circuit that form the arithmetic sum of three input bits. **5**

10. a) Construct a 4-to-1 line multiplexer using logic gates. Explain its working procedure. **5**

    b) Implement the following Boolean function using 8 : 1 multiplexer. **5**

    $F(A, B, C, D) = \sum m(1, 3, 5, 6)$

11. Discuss any two addressing modes of any generic microprocessor with an example for each. **10**

12. Explain distributed memory MIMD architecture with neat diagram. **10**

13. Explain three state – buffer with all supporting diagrams. **10**

14. What is multithreading with respect to any generic microprocessor / OS ? Discuss multithreaded processor architecture in detail with neat diagram. **10**

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I Semester M.C.A. (2 Years Course) Degree Examination, July 2022
(CBCS)
(2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA3 : Computer Organization and Architecture

Time : 3 Hours Max. Marks : 70

***Instructions** : Answer **any five** from Part – **A** and **any four** from Part – **B**.*

PART – A

Answer **any five** of the following. **Each** question carry **six** marks. **(5×6=30)**

1. Convert the following numbers from decimal to octal and then to binary. Compare the binary numbers obtained with the binary numbers obtained directly from the decimal numbers.

   a) 375 b) 249

2. Explain full Adder functionality with suitable diagram.
3. Explain RS Flip Flops.
4. Compare and contrast RISC Vs CISC.
5. Discuss any three addressing modes in detail.
6. Discuss in detail memory mapped I/O Vs isolated I/O.
7. Discuss the role of DMA.
8. Explain distributed memory MIMD architecture with neat diagram.

PART – B

Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9. a) Perform the following subtractions using 1's and 2's complement method. **5**

      i) 01100 – 0001

      ii) 0011.1001 – 0001.1110

   b) Explain K-map explanation technique and simplify the following Boolean expression with neat diagram.

      $\Sigma(X, Y, Z) = \Sigma(1, 3, 5, 6, 8, 9)$. **5**

P.T.O.

10. Explain Multiplexer and De-multiplexer functionality with suitable diagram. **10**

11. Explain in detail the instruction formats and three-address, two-address, one-address and zero-address instructions with examples. **10**

12. Explain the memory reference instructions with the help of a flow chart. **10**

13. Discuss input-output instructions and program interrupt with the help of a flow chart. **10**

14. Write short notes on : **(5+5)**

    a) Types of ROM

    b) Inter Processor Communication

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COMPUTER SCIENCE
1MCA3 : Computer Organization and Architecture

Time : 3 Hours Max. Marks : 70

Instruction : *Answer* ***any five*** *questions from Section –* ***A*** *and* ***any four*** *questions from Section –* ***B****.*

SECTION – A

Answer **any five** questions. **Each** carries **six** marks. **(5×6=30)**

1. Explain error detection with odd parity bit and even parity bit.
2. What is an adder ? Give logic diagram and truth table of half adder circuit.
3. Write the differences between CISC and RISC architecture.
4. Discuss about memory reference instructions with suitable example.
5. Explain priority interrupt and discuss any one method with neat diagram.
6. Discuss the different mapping techniques used in cache memories.
7. Explain interrupt driven input output techniques.
8. How parallel operations are implemented effectively in computer system ?

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SECTION – B

Answer **any four** questions. **Each** carries **ten** marks. **(4×10=40)**

9. a) Explain the bus structures in a computer system with a neat diagram. **5**
   b) Distinguish between multiprocessors and multi computers. **5**
10. a) Perform $(1101)_2 - (0011)_2$ using 2's complement. **3**
    b) Convert the following $(1010.01)_2$ to $(\ ?\ )_{10}$ **3**
    c) Prove NAND and NOR are universal gates. **4**

P.T.O.

11. a) Briefly explain the different instruction formats with suitable examples. 5

    b) Discuss about steps involved in instruction cycle with interrupt enabled. 5

12. a) Explain the basic organization of a micro programmed control unit. 5

    b) Describe the data transfer method using DMA. 5

13. What are addressing modes ? Explain the various addressing modes with examples. 10

14. a) What is cache coherence problem ? Discuss about different cache coherence approaches. 5

    b) Write short notes on interprocessor communication and synchronization. 5

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I Semester M.C.A. (Two Years Course) Degree Examination, June/July 2024
(CBCS) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA3 : Computer Organisation and Architecture

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **any five** questions from Part – **A**.* ***(5×6=30)***
*2) Answer **any four** questions from Part – **B**.* ***(4×10=40)***

PART – A

I. Answer **any five** questions : **(5×6=30)**

1) Convert the following :
   i) $F3A7_{(16)}$ = _____ (8)
   ii) $204.125_{(10)}$ = _____ (16)
   iii) $11111110_{(2)}$ = _____ (10)
2) Explain addressing models and its types.
3) With a neat diagram explain Von-Neuman architecture.
4) Write a detailed note on different characteristics of multiprocessor.
5) Explain instruction level parallelism in detail.
6) Differentiate between CISC and RISC.
7) Explain different types of ROM.
8) What are the key architectural components and design principles employed in the development of a multi-threaded processor ? Explain in detail.

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PART – B

II. Answer **any four full** questions. **(4×10=40)**

9) a) With a neat circuit diagram and truth table, explain the working of full adder. **5**
   b) Analyse booth multiplication algorithm with flow chart, hardware implementation with an example. **5**

P.T.O.

10) a) Construct a 4 to 1 line multiplexer using logic gates. Explain its working procedure. **5**

b) Explain error detection using Hamming code. **5**

11) a) With a neat circuit diagram explain the working of JK flip-flop. **5**

b) Explain interrupt cycle with flow chart. **5**

12) a) What is memory-mapped I/O and program controlled I/O ? **5**

b) Explain different instruction formats with an example for each. **5**

13) Explain distributed memory MIMD architecture with a neat diagram. **10**

14) What are interconnection structures ? Explain interprocessor communication and synchronization. **10**

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I Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA3 : Computer Organisation and Architecture

Time : 3 Hours | Max. Marks : 70

Instructions : *1) Answer **any 5** questions from Part – **A**.*
*2) Answer **any 4** questions from Part – **B**.*

PART – A

I. Answer **any five** questions. **(5×6=30)**

1) Convert the following :
   i) $ACF_{(16)}$ = _____ (10)
   ii) $304.25_{(10)}$ = _____ (2)
   iii) $101101.11_{(2)}$ = _____ (10).
2) Write the differences between Half Adder and Full Adder.
3) Explain the working of master slave flip-flop.
4) Explain different instruction formats with an example for each.
5) Describe data transfer methods using DMA.
6) Briefly explain memory hierarchy.
7) What is memory mapped I/O and program controlled I/O ?
8) Explain different types of ROM.

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PART – B

II. Answer **any 4 full** questions. **(4×10=40)**

9) a) Minimize the following expression
   $F(A, B, C, D) = \sum(5, 6, 9, 13, 15) + d(1, 7, 14)$. **5**
   b) Explain error detection using hamming code. **5**
10) a) Implement the following boolean function using 8 : 1 multiplexer
   $F(A, B, C, D) = \sum m(0, 1, 5, 6, 7, 9, 10, 15)$. **5**
   b) Explain instruction level parallelism in detail. **5**

P.T.O.

11) a) Differentiate between RISC and CISC. 5

b) Content of a 4-Bit register is initially 1101. Register is shifted six times to the right with serial input 101101. What is the content of the register after each shift ? 5

12) a) What is cache coherence problem ? Discuss different cache coherence approaches. 5

b) Explain the basic organization of microprogrammed control unit. 5

13) Explain distributed memory MIMD architecture with a neat diagram. 10

14) Briefly explain inter process communication and process synchronization. 10

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